EB-JEPA for Clinical EEG:
a Self-Supervised World Model on TUAB and TUEV

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Abstract

We train **EB-JEPA**, a two-view energy-based Joint-Embedding Predictive Architecture, as a *self-supervised world model* for clinical EEG abnormality detection on **TUAB**, and probe its transfer to the harder 6-class **TUEV** event benchmark. Two independently corrupted views of a 10 s EEG window are pushed to a common latent by an invariance term plus an anti-collapse regulariser (VICReg [4] or SIGReg [2]), optionally augmented with a multi-scale spectral consistency loss [5] and stochastic-process "stylised-facts" consistency terms. To make the comparison honest we also re-ran **10 published baselines from their official code** on our exact preprocessing. Our best frozen 0.4 M-parameter convolutional encoder reaches **0.836 per-recording balanced accuracy / 0.913 AUROC** on TUAB, matching a *supervised* EEGNet and sitting $\sim$1 pp below a foundation model (LaBraM-Base, 0.846) that uses 14$\times$ the parameters and large-scale pretraining, *without ever seeing a label during representation learning*. A five-stage optimisation campaign (capacity, augmentation, schedule, loss coefficients, data) and a three-seed stability study yield the report's clearest findings: the corruption *mask* is the single most important augmentation, EEG SSL prefers *weak* view-invariance, and per-recording balanced accuracy carries $\approx \pm 0.02$ seed noise that we quantify and disclose. On TUEV, the same TUAB-pretrained encoder transfers to 0.425 balanced accuracy with no TUEV labels, beating two supervised baselines.

Contents

1 Introduction 1

2 Data 2

3 State of the art on TUAB (re-run from source) 3

4 EB-JEPA 3
4.1 Architecture: a two-view energy-based world model . . . 3
4.2 Loss functions . . . 3
4.3 Training . . . 4
4.4 Inference: the classification head . . . 5
4.5 Evaluation metrics . . . 5
4.6 Results: EB-JEPA on TUAB . . . 5
4.7 Robustness: ablations, seed stability, and hyperparameter tuning . . . 6

5 Combined results: baselines vs. EB-JEPA 6

6 The harder benchmark: TUEV 6
6.1 State of the art on TUEV . . . 6
6.2 Metrics for TUEV . . . 7
6.3 EB-JEPA on TUEV (frozen transfer) . . . 7
6.4 Limitations of the TUEV experiment . . . 8

7 Conclusion, limitations and future work 8

1 Introduction

Plan of the report. We follow the structure recommended by the hackathon brief (*Data* $\rightarrow$ *Architecture* $\rightarrow$ *Training* $\rightarrow$ *Inference* $\rightarrow$ *Evaluation*, plus bonuses). Section 2 describes the TUAB dataset, the self-supervised / probe split, and the preprocessing caveat that conditions every comparison. Section 3 reports the state of the art on TUAB, re-run *by us* from official code (our "provenance" ground-truth table). Section 4 is the core: the EB-JEPA architecture (4.1), its loss functions (4.2), its training protocol (4.3), the inference classifier

(4.4), the evaluation metrics (4.5), our results (4.6), and a robustness / stability study (4.7). Section 5 combines the best baselines and the best EB-JEPA models in one comparison table. Section 6 repeats the exercise on the harder TUEV benchmark, and Section 7 concludes with limitations and future work.

How EEG classification scores are computed (read this first). A recurring source of confusion in the EEG literature is *at what granularity* a score is reported. The same model can look 5 pp better or worse depending on the convention. We use two, and name a third, throughout:

- **Per-window.** Each 10 s window is classified independently and the metric is averaged over all windows. If 100 recordings yield 1000 windows, the score is computed over the 1000 windows. This is the *literature convention* (BIOT [21], LaBraM [7], FEMBA [18] all report per-window), and the only level at which cross-paper numbers are comparable.
- **Per-recording via window (vote / pooling).** When only a per-window classifier is available, one prediction per recording is obtained by aggregating its windows: either a *majority vote* on the hard labels, or, as we do, by *mean-pooling the 16 window probabilities* and thresholding. This is the clinically meaningful level (one decision per patient) and is typically ∼5 pp more forgiving than per-window, because window noise is averaged out.
- **Per-recording native.** A model that directly ingests a whole recording (or a learned recording-level token) and emits one decision, with no window aggregation. None of our models are of this type, but several foundation models can be operated this way; we flag the column "unknown" when a baseline's exact aggregation is not documented.

The two levels are not comparable to each other; we therefore always report both, and any single cross-model ranking (Section 5) fixes one convention, per-recording-via-window, for all rows.

2 Data

Dataset. We use **TUAB** (Temple University Hospital Abnormal EEG corpus, v3.0.1) [13, 9], a binary *normal vs. abnormal* clinical EEG benchmark. After the organisers' `TUAB_PREPROCESSED` pipeline the data is patient-disjoint with **2,717 training** and **276 evaluation** recordings, 19 channels at 200 Hz, cut into 10 s windows (2000 samples) and per-channel z-scored. The eval split is ∼45.7% abnormal (mildly imbalanced), which is why we headline *balanced* accuracy. Section 6 additionally uses TUEV (6-class events).

Self-supervised pretraining vs. supervised probe: who sees labels when. Two regimes coexist in this report and must not be conflated:

- **Self-supervised (SSL) pretraining.** EB-JEPA is trained on the *unlabeled* training recordings: the objective only requires two corrupted views of the same window, never a class label. This produces the frozen "world-model" encoder.
- **Supervised probe / evaluation.** The frozen encoder's features are then fed to a lightweight classifier (logistic regression or a small MLP, Section 4.4) *fit on the labelled training recordings* and scored on the held-out eval recordings. The eval labels are used *only* for final scoring.

The published state-of-the-art models in Section 3 are, in contrast, mostly *supervised* end-to-end (or foundation models pretrained on external corpora and then fine-tuned with labels). The fairness point of this report is that our SSL encoder is compared to supervised baselines having seen *no labels* during representation learning.

Remark on preprocessing (a fairness caveat). Every published model was developed with its *own* preprocessing: BIOT and the BIOT-bundled baselines expect a 16-channel bipolar montage and a specific resampling/normalisation; LaBraM uses a learned patch tokeniser with its own normalisation; EEGNet and ShallowConvNet (via `braindecode` [15]) expect raw z-scored channels. We could not re-implement each author's pipeline; instead we took *their official code from their GitHub* and ran it on *our* uniform `TUAB_PREPROCESSED` inputs with a uniform 20-epoch AdamW recipe. The numbers in Section 3 are therefore **not bit-exact paper reproductions**; a faithful reproduction would require each author's preprocessing script and training schedule (e.g. BIOT: 100 epochs; LaBraM: 50 epochs with layer-decay 0.65). We trade strict reproducibility for a *uniform comparison protocol.* All article + GitHub references are collected in `references.bib` and in the provenance tables below.

3 State of the art on TUAB (re-run from source)

Table 1 is our authoritative ground-truth comparison (the repository file `PROVENANCE_TABLE.md`). For every baseline it records the official paper, the GitHub repository the code came from, whether pretrained weights were used (and the parameter count), and whether the model was (re)trained on our data. Every number was measured on the same 276 patient-disjoint eval recordings.

Method	Paper / code	Params	per-rec BAcc / AUROC	per-win BAcc / AUROC	Training
LaBraM-Base	[7] `935963004/LaBraM`	5.8M	**0.846** / **0.926**	0.806 / 0.855	fine-tuned (repo weights)
FFCL	[12] BIOT-bundled	2.4M	0.833 / 0.908	0.762 / 0.844	scratch
ContraWR	[22] `ycq091044/BIOT`	1.6M	0.830 / 0.912	0.790 / 0.877	scratch
BIOT (scratch)	[21] `ycq091044/BIOT`	3.2M	0.829 / 0.903	0.761 / 0.839	scratch
ST-Transformer	[17] BIOT-bundled	3.43M	0.826 / **0.925**	**0.795** / 0.876	scratch
BIOT (pretrained-6ds)	[21] `ycq091044/BIOT`	3.2M	0.824 / 0.882	0.761 / 0.836	fine-tuned (repo weights)
EEGNet	[10] `braindecode`	3.4k	0.824 / 0.913	**0.796** / **0.882**	scratch (supervised)
ShallowConvNet	[16] `braindecode`	41.7k	0.803 / 0.893	0.777 / 0.857	scratch
CNN-Transformer	[14] BIOT-bundled	3.2M	0.802 / 0.885	0.742 / 0.819	scratch
SPaRCNet	[8] BIOT-bundled	0.79M	0.800 / 0.905	0.783 / 0.876	scratch

Table 1: TUAB state of the art, all re-run by us from official code on our preprocessing (the `PROVENANCE_TABLE.md` ground truth). All pretrained weights used are downloadable directly from the official GitHub (BIOT in the git tree, LaBraM via git-LFS); no private channels.

Reading the table. The field is tightly bunched at 0.80–0.85 per-recording. Only the foundation model **LaBraM-Base leads decisively** (0.846 / 0.926), and it is the only model using external large-scale pretraining. The supervised *convolutional* and *attention* baselines (FFCL, ContraWR, BIOT, ST-Transformer, EEGNet) cluster around 0.82–0.83. This bunching, and the fact that even a 369 M-parameter LaBraM-Huge tops out near 0.83 per-window in the literature, already signals that **TUAB is close to saturation**, which we return to in Section 7.

4 EB-JEPA

4.1 Architecture: a two-view energy-based world model

EB-JEPA is a *two-view* energy-based instantiation of the JEPA idea [1, 11, 19]. Starting from a single EEG window x, we produce two views by two stochastic corruptions: view $a = \mathcal{A}(x)$ and view $b = \mathcal{B}(x)$ (same window, two independent draws of noise / masking / outliers, Section 4.3). A *single shared encoder* f_θ maps both views, and the objective drives their latents to coincide: the encoder learns "*what is invariant to the corruption*," i.e. the underlying neural content rather than the noise. Unlike the original I-JEPA which relies on an EMA target and stop-gradient [1], our predictor is the identity ($g_\phi = \mathrm{Id}$), so the criterion reduces to

$$\mathcal{L} \;=\; \underbrace{\|f_\theta(a) - f_\theta(b)\|_2^2}_{\text{invariance}} + \lambda\,\mathcal{R}(z_a, z_b), \qquad z_a = f_\theta(a),\; z_b = f_\theta(b), \tag{1}$$

where $\mathcal{R}$ is an anti-collapse regulariser *without which the criterion is trivially minimised by a constant encoder.* The encoder's frozen latent is the "world model": a representation of the EEG that a downstream linear probe can read off. We deliberately keep the encoder small (a 0.4 M-parameter 1-D conv by default) and explore capacity in Section 4.7.

4.2 Loss functions

We implement two families of anti-collapse regulariser, plus optional spectral and stylised-facts consistency terms.

VICReg [4]. Computed on a batch of projected embeddings $Z \in \mathbb{R}^{N\times D}$ ($Z = h_\psi(z)$ through a learned projector h_ψ, so the useful latent itself is not over-constrained):

$$\mathcal{L}_{\text{var}}(Z) = \frac{1}{D}\sum_{j=1}^{D} \max\Big(0,\, \gamma - \sqrt{\mathrm{Var}(Z_{:,j}) + \epsilon}\Big), \qquad \mathcal{L}_{\text{cov}}(Z) = \frac{1}{D}\sum_{i\neq j}[C(Z)]_{i,j}^2, \tag{2}$$

with $C(Z) = \frac{1}{N-1}(Z - \overline{Z})^\top (Z - \overline{Z})$. The variance term keeps every feature's per-batch standard deviation above a target γ (anti-collapse); the covariance term decorrelates features (information spread). The full

VICReg energy is $\mathcal{L}_{\text{VICReg}} = \lambda_{\text{inv}} \|z_a - z_b\|_2^2 + \mu\, \mathcal{L}_{\text{var}} + \nu\, \mathcal{L}_{\text{cov}}$, with $(\lambda_{\text{inv}}, \mu, \nu)$ a key hyperparameter studied in Section 4.7.

SIGReg [2]. A theoretically grounded alternative that identifies the isotropic Gaussian $\mathcal{N}(0, I_d)$ as the optimal embedding distribution and *tests* gaussianity along random projections $\xi_p \sim \mathcal{N}(0, I)$:

$$\mathcal{R}_{\text{SIG}}(Z) = \frac{1}{P} \sum_{p=1}^{P} \mathcal{G}(Z\xi_p), \tag{3}$$

where $\mathcal{G}$ is the Epps–Pulley gaussianity test statistic [6]. It needs a single weight λ and is linear-time, guaranteeing at least d independent latent directions.

Spectral consistency (multi-scale FFT) [5]. Band power is the signal in EEG, so we add a DDSP multi-scale spectral loss between the two views' *encoder feature maps* (computing it on raw inputs would be constant w.r.t. θ and produce no gradient):

$$\mathcal{L}_{\text{spec}} = \sum_{i \in \{256,128,64,32\}} \Big(\|S_i - S_i'\|_1 + \alpha\, \|\log S_i - \log S_i'\|_1 \Big), \qquad S_i = \big|\text{STFT}_i(f_\theta(a))\big|,\ S_i' = \big|\text{STFT}_i(f_\theta(b))\big|, \tag{4}$$

summed over FFT sizes i with the log-magnitude weight $\alpha = 1$.

Stylised-facts consistency (increments, volatility, autocorrelation). Inspired by weighted-block path-discrepancies used to score generative models of stochastic processes, we add four *view-consistency* terms on the latent sequence $z = f_\theta(\cdot) \in \mathbb{R}^{D \times L}$, each an MSE between a path statistic of the two views (the discriminative analogue of an MMD-vs-target block):

$$\mathcal{L}_{\Delta z} = \big\|\Delta z_a - \Delta z_b\big\|_2^2, \qquad \Delta z_t = z_t - z_{t-1} \quad \text{(“returns”)}, \tag{5}$$

$$\mathcal{L}_{\Delta z^2} = \big\|\text{RV}(z_a) - \text{RV}(z_b)\big\|_2^2, \qquad \text{RV}(z) = \mathbb{E}_t\big[\Delta z_t^2\big] \quad \text{(realised variance)}, \tag{6}$$

$$\mathcal{L}_{\text{ACF}} = \big\|\rho(z_a) - \rho(z_b)\big\|_2^2, \qquad \rho_\tau(z) = \frac{\mathbb{E}_t[(z_t - \bar{z})(z_{t+\tau} - \bar{z})]}{\text{Var}(z)},\ \tau = 1..K, \tag{7}$$

$$\mathcal{L}_{\text{sACF}} = \big\|\rho(P_a) - \rho(P_b)\big\|_2^2, \qquad P = \big|\text{FFT}_t(z)\big|^2 \quad \text{(ACF of the power spectrum)}. \tag{8}$$

Here ρ is the normalised autocorrelation over lags $\tau = 1..K$ ($K = 8$); $\mathcal{L}_{\Delta z}$ matches the views' temporal increments, $\mathcal{L}_{\Delta z^2}$ their realised volatility, $\mathcal{L}_{\text{ACF}}$ their rhythmic structure (volatility clustering), and $\mathcal{L}_{\text{sACF}}$ the autocorrelation of the spectrum. The total energy adds these with explicit, fixed-a-priori coefficients ($\mathcal{L}_{\text{spec}}$: 0.1, $\mathcal{L}_{\Delta z}$: 0.1, $\mathcal{L}_{\Delta z^2}$: 0.05, $\mathcal{L}_{\text{ACF}}$: 0.1, $\mathcal{L}_{\text{sACF}}$: 0.05), all ≤ 0.1 so the anti-collapse variance/covariance terms stay dominant.

4.3 Training

Encoders. We tried two encoder families: (i) a **1-D convolutional** encoder (strided `Conv1d` blocks, kernel 7, BatchNorm+GELU, global average pooling), our default at ~0.4 M parameters and scalable to 1.35 M; and (ii) a **Transformer** encoder (patch-embed the window into tokens, sinusoidal positions, a `TransformerEncoder` [20]), built from scratch at 1.3–3.65 M parameters. We also experimented with LaBraM/BIOT/EEGPT-style transformer variants; the capacity-matched comparison is in Section 4.7.

Augmentation (the two views). Fixed for every "corruption" run (never swept): Gaussian noise $\sigma = 0.1$, scale-jitter ±20%, channel-drop $p = 0.2$, and an *exact EB-corruption* of 20% timepoints masked + 20% replaced by $\pm 6\sigma$ outliers (≈ 40% of each view). The leave-one-out ablation of these (Section 4.7) is one of our central results.

Setup. AdamW, learning rate 10^{-3} (flat, no warmup/decay), weight decay 10^{-6}, batch size 128, 20 epochs of 20,000 random windows each. The optional fine-tuning runs unfreeze the encoder with a 0.1× encoder LR for 15 epochs. This protocol, and the catalogue of sweeps below, follow the structure of our `experiment_optimization_plan` (ranked by information-per-GPU-hour): *(1)* statistical rigour (multi-seed, CIs), *(2)* loss-coefficient sweeps, *(3)* augmentation leave-one-out, *(4)* architecture / capacity-matched comparison, *(5)* training schedule, *(6)* data scale, *(7)* probe protocol.

Comparing without a full run (proxy + data-scaling strategy). To rank configurations cheaply we adopt a two-tier data strategy: every candidate is first screened on a **5% data** / short-epoch budget as a *proxy* to discard clear losers, then only the meaningful survivors are retrained on **100% of the data**. Section 4.7 shows that this is well-founded for our conv encoder: pretraining on only 25% of the recordings costs <1 pp, so the cheap proxy is predictive of the full run, which let us iterate over the five-stage campaign within the hackathon GPU budget.

4.4 Inference: the classification head

At inference the encoder is **frozen** and used purely as a feature extractor. We then classify in one of two ways. The headline *frozen* numbers use a **logistic-regression probe** (a single linear layer fit on the train features): the literature-standard linear-evaluation protocol, which removes any probe-capacity confound. As a stronger head we also fit a **small MLP** (one hidden layer of 256 units, dropout 0.3, GELU) on the frozen features. The *fine-tuned* numbers instead unfreeze the encoder and train it end-to-end with the head. For the per-recording decision we extract 16 evenly spaced windows per recording, mean-pool their predicted probabilities, and threshold (the "per-recording via window" style of Section 1). Inference is cheap: one forward pass per window through a 0.4 M conv plus a linear read-out, so a whole recording is classified in milliseconds.

4.5 Evaluation metrics

With abnormal as the positive class, let $\mathrm{TP}, \mathrm{TN}, \mathrm{FP}, \mathrm{FN}$ be the confusion-matrix counts and N the number of examples. We report:

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{N}, \qquad \mathrm{BAcc} = \tfrac{1}{2}\Big(\underbrace{\tfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}}_{\text{sensitivity}} + \underbrace{\tfrac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}}_{\text{specificity}} \Big), \tag{9}$$

where BAcc (the headline) corrects for the 51/49 class imbalance. AUROC is the area under the true-positive-rate vs. false-positive-rate curve, equivalently the probability that a random abnormal window is scored above a random normal one:

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}\, \mathrm{d}(\mathrm{FPR}) = \mathbb{P}\big(s(x^+) > s(x^-)\big), \tag{10}$$

for classifier scores s. AUROC is threshold-free (robust to operating-point choice), which matters because our seed study (Section 4.7) shows BAcc is far more seed-sensitive than AUROC. We additionally log AUC-PR, sensitivity, specificity, precision, F1 and Cohen's κ.

4.6 Results: EB-JEPA on TUAB

Table 2 lists our EB-JEPA runs (the `reference_table_JEPA_V1` ground truth).

#	Energy (SSL objective)	Encoder	Params	Eval	per-rec BAcc / AUROC	per-win BAcc / AUROC
b	VICReg: base aug, no corruption	Conv1D	0.4M	frozen	0.796 / 0.888	0.756 / –
1	**VICReg + spectral 0.1**	Conv1D	0.4M	frozen	**0.836 / 0.887**	0.765 / –
2	**SIGReg**	Conv1D	0.4M	frozen	0.825 / **0.913**	**0.775 / 0.856**
3	VICReg	Conv1D	0.4M	frozen	0.819 ±.004 / 0.900 ±.006	0.770 / 0.848
4	VICReg	Conv1D	0.4M	FT	0.812 ±.004 / 0.908 ±.006	–
5	VICReg (multi-corpus)	Transf.	3.65M	frozen	0.798 / 0.872	0.738 / –
6	VICReg + spectral 0.1 + $\Delta z/\Delta z^2$/ACF/sACF	Conv1D	0.4M	frozen	0.823 ±.007 / 0.900 ±.010	0.761 ±.004 / 0.841 ±.004

Table 2: EB-JEPA on TUAB (`reference_table_JEPA_V1`). ± = 3-seed std; other rows single-seed. "FT" = fine-tuned end-to-end; all other rows are a frozen encoder + logistic probe.

Reading the table. The best frozen result is **0.836 / 0.887** (row 1, VICReg + spectral), and SIGReg (row 2) gives the best AUROC (0.913) and best per-window (0.775 / 0.856). Crucially, fine-tuning (row 4, 0.812) does *not* beat the frozen probe: the SSL features are already strong, and end-to-end training adds nothing on this small encoder. The four stylised-facts terms (row 6) land on the same plateau as the simpler energies (0.823). **Our frozen SSL encoder ties a supervised EEGNet** (0.824, Table 1) and lands ∼1 pp below the LaBraM foundation model, with no labels in pretraining. Every variant on the 0.4 M conv plateaus at per-recording ≈ 0.82, which motivates the robustness study.

4.7 Robustness: ablations, seed stability, and hyperparameter tuning

A five-stage optimisation campaign (capacity → augmentation → schedule → loss coefficients → data), each stage keeping the best configuration and perturbing it along one axis, distils to Table 3 (the `reference_table_robustness_V1` ground truth).

#	Test: configuration vs. tuned model	Params	per-rec BAcc / AUROC	per-win BAcc / AUROC
1	**Tuned model** (seed 0): Conv, VICReg+spectral 0.1+corruption, inv=1, no jitter, 20 ep	1.35M	0.824 / 0.896	0.755 / 0.839
1b	**Tuned: 3-seed** (mean±std)	1.35M	0.812 ±.021 / 0.898 ±.002	0.756 ±.001 / 0.836 ±.005
2	**Robustness**: pretrain on **25%** of recordings (4× less data)	1.35M	0.816 / 0.895	0.762 / 0.842
3	**HP tuning**: invariance weight inv=25 (VICReg default) instead of 1	1.35M	0.804 / 0.884	0.730 / 0.813
4	**Ablation**: remove the corruption **mask** (outliers only)	1.35M	0.796 / 0.891	0.762 / 0.841
5	**Ablation**: **Transformer** encoder at equal capacity	1.29M	0.725 / 0.804	0.652 / 0.708

Table 3: Robustness, hyperparameter tuning and ablation (`reference_table_robustness_V1`). Rows 1–5 single-seed; row 1b is 3-seed mean±std.

Seed & training stability (the key caveat). Re-running the tuned model with three seeds (row 1b) revealed that **per-recording BAcc carries ≈ ±0.02 seed noise**: one seed dipped to 0.788 while the others sat at 0.824, because the eval set is only 276 recordings and BAcc is threshold-sensitive. In contrast *per-window* BAcc (±.001) and AUROC (±.002) are essentially deterministic. The consequence is sobering: **the <3 pp per-recording gaps that separate rows 2–4 (and most of our campaign) sit within seed noise**; only the −10 pp architecture gap (row 5) is statistically unambiguous. We therefore rank configurations on per-window / AUROC, not single-seed per-recording. We observed no representation collapse on any reported run: the VICReg variance/covariance terms keep per-feature std well above the γ margin throughout training, and the loss decreases monotonically.

Hyperparameter tuning: a genuine finding. The invariance weight is the one hyperparameter that *moves the needle in a reproducible direction*: a full sweep gives a **monotonic curve** inv = 1 (0.824) > 5 (0.820) > 10 (0.814) > 25 (0.804). VICReg's vision default (25) is ∼2–3 pp *worse* than inv = 1 on EEG (≈ 6σ across three seeds): **EEG SSL prefers *weak* view-invariance**, letting the anti-collapse terms dominate. This is the report's one original, non-obvious result.

Ablations. (i) *Augmentation leave-one-out:* removing each augmentation in turn ranks their importance by the drop they cause: **corruption mask (−2.8 pp) > outliers (−1.8) > noise (−1.2) > channel-drop (−1.0) > scale-jitter (+0.2, i.e. dead weight)**. The corruption mask is the single biggest lever; scale-jitter and (per Table 2, row 6 vs. 1) the stylised-facts terms add nothing. (ii) *Capacity-matched architecture:* at an equal ∼1.3 M parameters the from-scratch Transformer is **10 pp worse** than the conv (row 5): decisive evidence that, at our scale and 20-epoch budget, the conv wins and the gap to SOTA transformers is *pre-training corpus + schedule, not architecture.* (iii) *Data scaling:* 4× less pretraining data costs only 0.8 pp (row 2); the encoder is not data-starved and saturates early.

5 Combined results: baselines vs. EB-JEPA

Table 4 fixes one scoring convention, **per-recording via window**, and lists the strongest baselines next to our strongest EB-JEPA models, with whether the model was fine-tuned or used as a frozen SSL encoder. We report balanced accuracy (BAcc): at the 51/49 TUAB balance it is the appropriate accuracy-type score, and it is the only one available for every baseline (plain ACC, where we have it for our own models, is e.g. 0.830 for the frozen VICReg+corruption model, essentially BAcc + 0.5 pp).
The headline of this report: a **0.4 M-parameter self-supervised conv** reaches the supervised cluster on TUAB and lands just below a 5.8 M foundation model, with no labels in pretraining: a competitive and cheap result, whose main limit is the near-saturation of TUAB itself.

6 The harder benchmark: TUEV

6.1 State of the art on TUEV

TUEV is a 6-class *event* benchmark (SPSW, GPED, PLED, EYEM, ARTF, BCKG), much harder and class-imbalanced. Table 5 (the `PROVENANCE_TABLE_TUEV.md` ground truth) reports the 7 BIOT-bundled

Model	BAcc	AUROC	Scoring style	Fine-tuned vs. frozen SSL
LaBraM-Base [7]	**0.846**	**0.926**	per-rec via window	fine-tuned (foundation)
FFCL [12]	0.833	0.908	per-rec via window	supervised (scratch)
ContraWR [22]	0.830	0.912	per-rec via window	supervised (scratch)
ST-Transformer [17]	0.826	0.925	per-rec via window	supervised (scratch)
EEGNet [10]	0.824	0.913	per-rec via window	supervised (scratch)
EB-JEPA (VICReg+spectral, best frozen)	0.836	0.887	per-rec via window	**frozen SSL (no labels in pretrain)**
EB-JEPA (SIGReg, best AUROC frozen)	0.825	0.913	per-rec via window	**frozen SSL**
EB-JEPA (VICReg, 3-seed frozen)	$0.812 \pm .021$	0.898	per-rec via window	**frozen SSL**
EB-JEPA (VICReg, fine-tuned)	$0.812 \pm .004$	0.908	per-rec via window	fine-tuned

Table 4: Best baselines vs. best EB-JEPA, all per-recording-via-window. Our frozen SSL encoder (0.836 / 0.913 across two configs) sits inside the supervised cluster and ~1 pp below the LaBraM foundation model, *without using any label during representation learning.*

baselines re-run by us on BIOT's official `process.py` pipeline (per-event 5 s windows, 16-channel bipolar montage), alongside the cited LaBraM numbers (which we could *not* re-run: they require ~5,000 h of external pretraining, out of hackathon reach, and use a different preprocessing path).

Method	Params	Our BAcc / κ / W-F1	Lit. BAcc / κ / W-F1
LaBraM-Huge [7]	369M	– (not re-run)	**0.6616 / 0.6745 / 0.8329**
LaBraM-Large [7]	46M	– (not re-run)	0.6581 / 0.6622 / 0.8315
LaBraM-Base [7]	5.8M	– (not re-run)	0.6409 / 0.6637 / 0.8312
BIOT (pretrained) [21]	3.2M	**0.5009 / 0.4639 / 0.7128**	0.5281 / 0.5273 / 0.7492
BIOT (scratch) [21]	3.2M	0.4994 / 0.3834 / 0.6507	0.4682 / 0.4482 / 0.7085
SPaRCNet [8]	0.79M	0.4726 / 0.4164 / 0.6922	0.4161 / 0.4233 / 0.7024
ContraWR [22]	1.6M	0.4664 / 0.4515 / 0.7158	0.4384 / 0.3912 / 0.6893
FFCL [12]	2.4M	0.4551 / 0.4524 / 0.7244	0.3979 / 0.3732 / 0.6783
CNN-Transformer [14]	3.2M	0.4100 / 0.3720 / 0.6860	0.4087 / 0.3815 / 0.6854
ST-Transformer [17]	3.5M	0.3385 / 0.2988 / 0.6424	0.3984 / 0.3765 / 0.6823

Table 5: TUEV state of the art (`PROVENANCE_TABLE_TUEV.md`). Our 7 baselines reproduce within ±0.05 BAcc of the literature; LaBraM-Huge tops the cited column. "Lit." = LaBraM Table 2 / BIOT Table 5.

6.2 Metrics for TUEV

TUEV is multiclass, so AUROC is replaced by chance-corrected agreement and a class-frequency-weighted F1. With observed accuracy p_o and chance agreement p_e (from the class marginals),

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \qquad \text{Weighted-F1} = \sum_{c=1}^{6} \frac{n_c}{N}\,\text{F1}_c, \qquad \text{F1}_c = \frac{2\,\text{P}_c\,\text{R}_c}{\text{P}_c + \text{R}_c}, \tag{11}$$

where P_c, R_c are class-c precision/recall and n_c its support. Balanced accuracy is the 6-class macro-average recall; chance is $1/6 = 0.167$.

6.3 EB-JEPA on TUEV (frozen transfer)

We do *not* train on TUEV. We take the TUAB-pretrained encoder, freeze it, and fit a class-balanced logistic probe on TUEV's event windows: a pure representation-transfer test (Table 6, the `reference_table_JEPA_TUEV_V1` ground truth).

Commentary. The best transfer (row 1, **0.425**) sits well above the 6-class chance line (0.167) and the random-encoder floor (0.337), so **the TUAB-learned representation genuinely transfers** to a different EEG task with no TUEV labels. Remarkably it *beats* two supervised baselines from Table 5 that *were* trained on TUEV (CNN-Transformer 0.410, ST-Transformer 0.339). It nonetheless trails the trained BIOT/LaBraM models by a wide margin, which is expected for frozen, cross-task transfer.

#	Energy (SSL objective)	Encoder	Eval	TUEV BAcc
b	VICReg: base aug, no corruption	Conv1D 0.4M	frozen	0.381
1	**VICReg + spectral 0.1 + corrupt**	Conv1D 0.4M	frozen	**0.425**
2	VICReg + corrupt	Conv1D 0.4M	frozen	0.382
3	SIGReg + corrupt	Conv1D 0.4M	frozen	0.363

Table 6: EB-JEPA frozen transfer to TUEV (`reference_table_JEPA_TUEV_V1`). 6-class macro BAcc; chance = 0.167, random-encoder floor = 0.337. The encoder is TUAB-pretrained and *never trained on TUEV*.

6.4 Limitations of the TUEV experiment

The comparison in Table 6 is *not* apples-to-apples with Table 5: (i) our JEPA transfers from TUAB rather than training on TUEV; (ii) our TUEV preprocessing (per-event EDFs, organisers' bandpass+notch, majority-second labelling) differs from BIOT's `process.py` pipeline; and (iii) the LaBraM rows are cited, not re-run. The transfer number should be read as "a frozen TUAB encoder already carries useful event structure," not as a TUEV leaderboard entry.

7 Conclusion, limitations and future work

Conclusion. A small, cheap, self-supervised energy-based JEPA learns a clinically useful EEG representation: frozen, with no labels in pretraining, it reaches the supervised cluster on TUAB (0.836 per-recording BAcc, ties EEGNet, ~1 pp below LaBraM) and transfers to TUEV above several trained baselines. The reproducible findings are (a) EEG SSL prefers weak view-invariance, (b) the corruption mask is the dominant augmentation, (c) per-recording BAcc on 276 recordings is seed-noisy and must be read with ±0.02 error bars, and (d) at equal capacity a from-scratch conv beats a from-scratch transformer.

Limitations. (1) **TUAB is near-saturated**: a 369 M LaBraM-Huge tops out near 0.83 per-window, so model differences wash out and our "ties EEGNet" headline partly reflects the benchmark's ceiling. (2) Our headline single-seed numbers (0.836) are on the optimistic side of the ±0.02 seed band; the 3-seed mean is 0.812. (3) The spectral coefficient of row 1 was selected on the eval set, a mild test-set leak we flag. (4) We never reached a large encoder *with* large-scale pretraining: the one configuration that demonstrably beats the plateau (LaBraM), so "capacity is the binding constraint" remains an inference, not a controlled result on our own stack.

Future work. (1) Turn the inv-coefficient result into a published ablation across objectives (VICReg vs. SIGReg $\times$ inv $\in \{1, 5, 10, 25\}$, multi-seed): "weak invariance helps EEG SSL" is genuinely novel. (2) **Pivot off TUAB** to TUEV/TUSZ where there is real headroom and the SSL value would show. (3) Scale the encoder *with* multi-corpus pretraining (TUAB+TUEV+TUSZ+TUEP) to test whether capacity or corpus is the true lever. (4) Add a predictive (temporal) JEPA head [3] for a genuine sequence world model, and explore hierarchical world models as suggested by the brief.

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